THE PI-ADIC COMPLETION OF A LOCALIZATION OF AN INTEGRAL PERFECTOID RING IS AGAIN INTEGRAL PERFECTOID

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Allowed standard prerequisites (editorial).

- The definition of an integral perfectoid ring with pseudo-uniformizer pi and the attached formal scheme. Set up in the paragraph preceding the counted statement (source line 467). Fixes the two properties that have to be verified for the completed localization: pi-adic completeness with pi a non-zero-divisor, and the Frobenius being surjective on R/pi.
- The short exact sequences R/pi^n -> R/pi^{n+k} -> R/pi^k -> 0 coming from the fact that pi is not a zero divisor. Used inside the counted proof at lines 475-478. Supplies the sequences that are localized and then passed to the inverse limit.

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Lemma 0.1. *Let $f \in R$. Then $R\langle f^{-1}\rangle = \varprojlim_n R/\pi^n[f^{-1}]$ is an integral perfectoid ring.*

Proof. Let $n, k \geq 0$, as π is not a zero divisor we have a short exact sequence

$$0 \to R/\pi^n \xrightarrow{\pi^k} R/\pi^{n+k} \to R/\pi^k \to 0.$$

Localizing at f and taking inverse limits on n we obtain

$$0 \to R\langle f^{-1}\rangle \xrightarrow{\pi^k} R\langle f^{-1}\rangle \to R/\pi^k[f^{-1}] \to 0.$$

Then $R\langle f^{-1}\rangle$ is π-adically complete and π is not a zero-divisor. On the other hand, localizing at f the Frobenius map $\varphi : R/\pi \xrightarrow{\sim} R/\pi^p$ one gets

$$\varphi : R/\pi[f^{-1}] \xrightarrow{\sim} R/\pi^p[f^{-p}] = R/\pi^p[f^{-1}],$$

which proves that $R\langle f^{-1}\rangle$ is an integral perfectoid ring. □

References

[1] R. Hartshorne - Residues and duality , Lect. Notes in Math. , vol. 20 , Springer-Verlag, Berlin-New York, 1966

[2] D. Mumford - Abelian varieties , TIFR Studies in Mathematics , vol. 5 , Tata Institute of Fundamental Research, Bombay, 2008